

PRINCE CHAUHAN

Senior Quantitative Researcher | Options, Volatility & Systematic Trading

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github.com/prince-chauhan-pc/quant-research-portfolio | SEBI Registered Research Analyst

PROFESSIONAL SUMMARY

Derivatives research professional with 9+ years of experience spanning options trading, market analysis, and systematic strategy development. Combines deep institutional derivatives expertise with AI-augmented quantitative engineering to design, prototype, and rigorously validate production alpha and execution pipelines across Indian equity derivatives and US index options. Experienced in strategy validation, portfolio risk monitoring, and automated research workflows.

PROFESSIONAL EXPERIENCE

SMC Global Securities Ltd. - Senior Quantitative Analyst & Researcher Dec 2024 - Present

- **Alpha Research Infrastructure:** Architected causal event-driven simulation pipelines and point-in-time alpha research on tick-level and 1-minute historical bar feeds across US and Indian index derivatives (SPX, VIX, Nifty, Bank Nifty).
- **US Index Derivatives:** Engineered automated execution pipelines and volatility surface modeling for short-dated US index options (SPX/SPXW) via Interactive Brokers (IBKR API), systematically capturing short-horizon IV skew mispricings with dynamic delta rebalancing.
- **Statistical Arbitrage:** Traded mean-reverting equity basket spreads across Bank Nifty constituents, maintaining strict delta neutrality via dynamic beta hedging and automated strike re-centering thresholds.
- **Walk-Forward Strategy Validation:** Delivered 2.5+ out-of-sample Sharpe and 2.1+ Profit Factor across a 7-year sample (2018-2024) of Nifty & Bank Nifty index options, evaluated net of 0.50% turnover friction.
- **TCA & Execution Optimization:** Reduced execution drag by 24% and eliminated broker order-state desynchronization via pre/post-trade Transaction Cost Analysis (TCA) telemetry and unique client order tracking.
- **Desk Mentorship & Risk Alignment:** Mentored 3 junior quantitative researchers and traders in statistical validation methods and execution best practices; partnered with Portfolio Managers on strategy scalability, real-time portfolio Greeks, dealer GEX mapping, parametric VaR, and SPAN margin caps.

Smart Bulls Capital - Research Analyst, Derivatives & Quantitative Research Mar 2017 - Nov 2024

- **Options Volatility Books:** Researched, structured, and traded systematic multi-leg options volatility strategies (Iron Condors, Butterflies, Calendar Spreads) with dynamic delta/vega rebalancing guided by real-time IV surface spline smoothing.
- **Implied Volatility & Greeks:** Inverted Black-Scholes prices numerically via Newton-Raphson and Brent algorithms; mapped dealer Gamma Exposure (GEX) to identify institutional pin-risk levels across weekly expiries.
- **Portfolio Construction & Allocation:** Implemented Risk Parity and Mean-Variance capital allocation models across multi-strike options books; managed daily P&L attribution, SPAN margin utilization, and drawdown limits within desk risk budgets.
- **Live Capital & Margin Scale:** Managed systematic intraday and multi-day options books deploying multi-crore margin; maintained strict SPAN margin utilization and contained drawdowns within desk risk tolerances.

Earlier Corporate Experience - Corporate Finance & Audit Roles 2014 - 2017

- **Corporate Treasury & Audit:** Executed treasury operations and working-capital forecasting at Indian Oil Corporation Ltd. (IOCL, 2016-2017); conducted statutory audits and corporate tax structuring at CA Matta & Associates (2014-2016).

SELECTED INDEPENDENT QUANTITATIVE RESEARCH (OPEN SOURCE)

Hosted at: github.com/prince-chauhan-pc/quant-research-portfolio

Point-in-Time Backtesting Framework & Invariant Gate (Python)

- **Research Objective:** Enforce point-in-time data alignment and realistic transaction friction in high-turnover options strategy research.
- **Methodology & Validation:** Implemented rolling walk-forward optimization partitioned into 70% in-sample parameter calibration, 20% out-of-sample cross-validation, and 10% held-out test windows to prevent data leakage, paired with 1,000-path Monte Carlo drawdown stress testing.
- **Audit Guardrails:** Engineered an automated data-integrity test suite enforcing Rules L1-L17 across 22 boundary checks, pre-deducting 0.50% round-trip premium friction and exchange SPAN margins.

Arbitrage-Free SVI Volatility Surface & GEX Profiling

- **Research Objective:** Eliminate calendar and butterfly arbitrage from market implied volatility quotes for robust cross-strike option pricing.
- **Methodology:** Calibrated daily implied volatility surfaces using SVI parameterization with non-linear optimization; enforced butterfly convexity and calendar-spread non-arbitrage constraints.
- **Practical Outcome:** Stabilized cross-strike smile interpolation across illiquid strikes and generated real-time dealer GEX sensitivity curves to map institutional pin-risk zones.

Bank Nifty Basket Cointegration & Statistical Arbitrage

- **Research Objective:** Systematically exploit stationary mean-reverting spreads among top banking constituent equities.
- **Methodology & Risk:** Applied Johansen multivariate cointegration tests to identify stationary linear combinations; estimated spread mean-reversion half-life to calibrate entry/exit thresholds and holding periods, maintaining delta neutrality via rolling OLS beta weights.

Microstructure Execution Simulator & TCA Engine

- **Research Objective:** Accurately quantify market crossing costs and adverse selection for algorithmic limit orders.
- **Methodology:** Modeled Limit Order Book (LOB) queue priority, non-linear market impact, and fill probability across market regimes classified by Parkinson realized volatility.

TECHNICAL & QUANTITATIVE COMPETENCIES

- **Quantitative Research:** Statistical Arbitrage, Johansen Cointegration, Dynamic Beta Hedging, SVI Volatility Surface Fitting, Parkinson Realized Volatility, Market Regime Classification (K-Means), Walk-Forward Optimization, Monte Carlo Simulation.
- **Derivatives & Risk:** Black-Scholes Pricing, Portfolio Greeks (Delta, Gamma, Vega, Theta), Dealer GEX Profiling, Parametric & Historical VaR, SPAN Cross-Margin Optimization, Risk Parity, Mean-Variance Allocation.
- **AI-Augmented Engineering:** Agentic LLM Architecture for Rapid Alpha Prototyping, Automated Pipeline Construction, Synthetic Scenario Generation, Code Verification, and Mathematical Invariant Auditing.
- **Core Languages & Systems:** Python (NumPy, SciPy, Polars), SQL (DuckDB, PostgreSQL), Apache Parquet, Git, Linux.
- **Markets & Microstructure:** Indian Equity Derivatives (NSE Nifty, Bank Nifty, Sensex), US Equity Derivatives (CBOE SPX, VIX), Pre/Post-Trade TCA, Limit Order Book Queue Dynamics, Smart Order Slicing.

EDUCATION & REGULATORY CREDENTIALS

- **Bachelor of Commerce (Honours):** Kirori Mal College, University of Delhi (2011 - 2014) | First Division (71% Marks)
- **Chartered Accountancy (CA Final):** Institute of Chartered Accountants of India (ICAI, 2018) | Cleared Group 1 (78% Marks)
- **Regulatory Licenses:** SEBI Registered Research Analyst | NISM Series-XV